Respiratory gases	Partial pressure mm Hg				
	Atmospheric air	Alveoli	Deoxygenated Blood	Oxygenated blood	Tissues
O_2	159	104	40	95	40
CO_2	0.3	40	45	40	45

Table 6.1 Partial pressure of Oxygen and Carbon dioxide (in mmHg) in comparison to those gases in the atmosphere

6.5 Transport of gases

6.5.1 Transport of oxygen

Molecular oxygen is carried in blood in two ways bound to haemoglobin within the red blood cells and dissolved in plasma. Oxygen is poorly soluble in water, so only 3% of the oxygen is transported in the dissolved form. 97% of oxygen binds with haemoglobin in a reversible manner to form oxyhaemoglobin (HbO_2). The rate at which haemoglobin binds with O_2 is regulated by the partial pressure of O_2. Each haemoglobin carries maximum of four molecules of oxygen. In the alveoli high pO_2, low pCO_2, low temperature and less H^+ concentration, favours the formation of oxyhaemoglobin, whereas in the tissues low pO2, high pCO_2, high H^+ and high temperature favours the dissociation of oxygen from oxyhaemoglobin.

Breathing through nose is healthier than through mouth– Why?

A **sigmoid curve** (S–shaped) is obtained when percentage saturation of haemoglobin with oxygen is plotted against pO_2. This curve is called **oxygenhaemoglobin dissociation curve** (Figure 6.7). This S–shaped curve has a steep slope for pO_2 values between 10 and 50mmHg and then flattens between 70 and 100 mm Hg.

Under normal physiological conditions, every 100mL of oxygenated blood can deliver about 5mL of O_2 to the tissues.

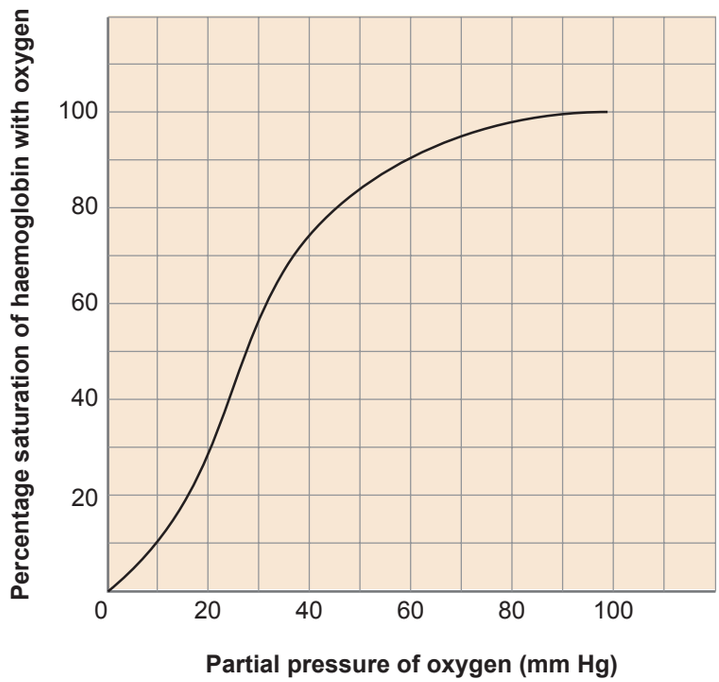


Figure 6.7 Oxygen dissociation curve

6.5.2 Transport of Carbon–dioxide

Blood transports CO_2 from the tissue cells to the lungs in three ways

XI Std Zoology Chapter-6 EM.indd 102 30/12/2022 10:41:57